

Императрица (Imperatriz) - Linguagem feita em Java para C

Instalação

Clonar o repositório do projeto

```
git clone https://github.com/PauloViniciusAF/Empress-Language.git
```

Como executar

```
./императрица <arquivo>.emp
```

Argumentos opcionais

```
--tree          //Retorna a AST no terminal
```

Glossário

- True: истинный
- False: ложь
- Integer: интеграл
- String: строка
- Decimal: десятичный
- Boolean: логический
- Do: делать
- While: пока
- For: для
- Function: функция
- Return: вернуть
- If: если
- Else: иначе
- Print: печать
- Input: входной
- List: список
- Continue: продолжать
- Break: перерыв

Expressões Regulares

- INT -> [0-9]+
- DEC -> [0-9]+.[0-9]+
- ID -> [a-zA-Z\u0040\u004FF0-9]+
- STRING -> "[a-zA-Z\u0040\u004FF_*/@=<>!\...]+"'
- BOOLEAN -> 'истинный | ложь'
- TYPE -> 'интеграл' | 'десятичный' | 'строка' | 'логический'

- PLUS -> '+'
- INCREMENT -> '++'
- MINUS -> '-'
- DECREMENT -> '--'
- TIMES -> '*'
- DIV -> '/'
- POW -> '^'
- MOD -> '%'
- COMMA -> ','
- SEMICOLON -> ';'
- ASSIGN -> ':'
- PLUS_ASSIGN -> '+:='
- MINUS_ASSIGN -> '-:='
- TIMES_ASSIGN -> '*:='
- POW_ASSIGN -> '^:='
- GREATER -> '>'
- LESS -> '<'
- GREATER_EQUAL -> '>='
- LESS_EQUAL -> '<='
- EQUAL -> '=='
- DIFFERENT -> '!='
- AND -> '&'
- OR -> '||'
- OPEN_BRACES -> '{'
- CLOSE_BRACES -> '}'
- OPEN_BRACKETS -> '['
- CLOSE_BRACKETS -> ']'
- OPEN_PARENTHESIS -> '('
- CLOSE_PARENTHESIS -> ')'
- OP_IF -> 'если'
- OP_ELSE -> 'иначе'
- OP_DO -> 'делать'
- OP_WHILE -> 'пока'
- OP_FOR -> 'для'
- OP_FUNCTION -> 'функция'
- OP_RETURN -> 'вернуть'
- OP_PRINT -> 'печать'
- OP_INPUT -> 'входной'
- OP_CONTINUE -> 'продолжать'
- OP_BREAK -> 'перерыв'
- EOF -> '\$'

Gramática

OBS: termos em caps lock são terminais, 'ε' representa produção vazia

file → bloco EOF

bloco → cmd bloco | ε

cmd → cmdIf | cmdFor | cmdWhile | cmdReturn | cmdDefFunc | cmdPrint | cmdInput | cmdID |
OP_CONTINUE | OP_BREAK

cmdID → tipo ID acessoListaOp complemento SEMICOLON | ID acessoListaOp complemento SEMICOLON

acessoListaOp → acessoLista acessoListaOp | ε

acessoLista → OPEN_BRACKETS expressaoAritmetica CLOSE_BRACKETS

complemento → ASSIGN valor | operadorAssignOp valor | OPEN_PARENTHESIS corpoLista
CLOSE_PARENTHESIS | ε

operadorAssignOp → PLUS_ASSIGN | MINUS_ASSIGN | TIMES_ASSIGN | DIV_ASSIGN | MOD_ASSIGN |
POW_ASSIGN

valor → expressaoLogica | lista | cmdPrint | cmdInput

expressaoLogica → expressaoRelacional (AND expressaoRelacional)* |
expressaoRelacional (OR expressaoRelacional)*

expressaoRelacional → expressaoAritmetica (opComparacao expressaoAritmetica)*

opComparacao → GREATER | LESS | EQUAL | DIFFERENT | GREATER_EQUAL | LESS_EQUAL

expressaoAritmetica → termo (opAdicao termo)*

opAdicao → PLUS | MINUS

termo → fator (opMultiplicacao fator)*

opMultiplicacao → TIMES | DIV | MOD

fator → elemento (POW elemento)*

elemento → INCREMENT ID X
| DECREMENT ID X
| ID X
| INT

| FLOAT
 | STR
 | BOOL
 | OPEN_PARENTHESIS expressaoLogica CLOSE_PARENTHESIS

$X \rightarrow composicao\ X \mid INCREMENT \mid DECREMENT \mid \epsilon$

$composicao \rightarrow OPEN_BRACKETS\ expressaoAritmetica\ CLOSE_BRACKETS\ acessoListaOp \mid OPEN_PARENTHESIS\ corpoLista\ CLOSE_PARENTHESIS$

$corpoLista \rightarrow valor\ entradaLista \mid \epsilon$

$entradaLista \rightarrow COMMA\ valor\ entradaLista \mid \epsilon$

$lista \rightarrow OPEN_BRACKETS\ corpoLista\ CLOSE_BRACKETS$

$cmdIf \rightarrow OP_IF\ valor\ OPEN_BRACES\ bloco\ CLOSE_BRACES\ cmdElse$

$cmdElse \rightarrow OP_ELSE\ OPEN_BRACES\ bloco\ CLOSE_BRACES \mid \epsilon$

$cmdWhile \rightarrow OP_WHILE\ valor\ OPEN_BRACES\ bloco\ CLOSE_BRACES$

$cmdFor \rightarrow OP_FOR\ OPEN_PARENTHESIS\ variavelFor\ SEMICOLON\ expressaoRelacional\ SEMICOLON\ expressaoAritmetica\ CLOSE_PARENTHESIS\ OPEN_BRACES\ bloco\ CLOSE_BRACES$

$variavelFor \rightarrow tipo\ ID\ complemento \mid ID\ complemento$

$cmdDefFunc \rightarrow OP_FUNCTION\ ID\ OPEN_PARENTHESIS\ listaParametros\ CLOSE_PARENTHESIS\ OPEN_BRACES\ bloco\ CLOSE_BRACES$

$listaParametros \rightarrow ID\ entradaListaParam \mid \epsilon$

$entradaListaParam \rightarrow COMMA\ ID\ entradaListaParam \mid \epsilon$

$cmdReturn \rightarrow OP_RETURN\ valorRetorno$

$valorRetorno \rightarrow valor \mid \epsilon$

$cmdPrint \rightarrow OP_PRINT\ OPEN_PARENTHESIS\ corpoLista\ CLOSE_PARENTHESIS\ SEMICOLON$

$cmdInput \rightarrow OP_INPUT\ OPEN_PARENTHESIS\ corpoLista\ CLOSE_PARENTHESIS\ SEMICOLON$

$tipo \rightarrow INT_TYPE \mid FLOAT_TYPE \mid BOOL_TYPE \mid STR_TYPE$

$id \rightarrow ID$